Syed Aayan Mahmood

AI Engineering — Backend Development

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Github | Linkedin | Verdict AI

Summary

Computer Science student at FAST NUCES with a strong interest in AI Engineering, Backend Development, and Large Language Models (LLMs). Experienced in developing AI-powered applications using FastAPI, LangGraph, React, PostgreSQL, Supabase, and vector databases. Passionate about building scalable multi-agent systems, AI evaluation platforms, and production-ready backend architectures.

Technical Skills

Programming Languages: Python, JavaScript, Java, C++, SQL
Backend Development: FastAPI, REST APIs, SQLAlchemy
AI & Machine Learning: LLMs, LangGraph, Agentic AI, RAG, Prompt Engineering, AI Evaluation, Gemini API
Frontend Development: React, HTML, CSS
Databases: PostgreSQL, Supabase, Qdrant, Vector Databases
Developer Tools: Git, GitHub, Docker, Linux, VS Code, PyCharm

Selected Projects

Verdict AI – AI-Powered LLM Evaluation Platform | Jul. 2026 – Present
Python, FastAPI, SQLAlchemy, LangGraph, React, PostgreSQL, Supabase, Gemini API
GitHub | Live Demo

- Built a React dashboard supporting response evaluation, model comparison, analytics, and prompt experiments.
- Designed a parallel LangGraph workflow with independent Accuracy, Logic, and Completeness evaluation agents followed by an adjudicator that generates a final verdict.
- Built a scalable FastAPI backend using PostgreSQL, Supabase, and SQLAlchemy for persistent storage of evaluations, prompt experiments, analytics, and evaluation history.
- Integrated JWT authentication and role-based dashboards for secure user management.

Aureon – Multi-Agent AI Orchestration Platform | Jul. 2026 – Present
Python, FastAPI, LangGraph, PostgreSQL, Docker, MCP

- Architecting a modular multi-agent AI orchestration platform for coordinating specialized AI agents across complex workflows.
- Implementing an orchestration engine supporting agent communication, workflow execution, task routing, and memory management.
- Designing scalable REST APIs using FastAPI while following a modular architecture designed for production deployment and extensibility.

PDF ChatBot – Retrieval-Augmented Generation (RAG) Assistant | Jun. 2026 – Jul. 2026
Python, FastAPI, Gemini API, Qdrant, RAG
GitHub

- Implemented semantic document retrieval using Qdrant vector database and Retrieval-Augmented Generation.
- Implemented document chunking, semantic search, embeddings, and vector retrieval with Qdrant for contextual question answering.
- Built a scalable FastAPI backend integrating Gemini API for conversational document analysis.

Mini Map – Intelligent Navigation & Route Analysis System | Sep. 2025 – Apr. 2026
C++, React, Leaflet, OpenStreetMap
GitHub

- Developed a Google Maps-inspired navigation system using OpenStreetMap data.
- Implemented Dijkstra's and A* algorithms for shortest-path computation and optimized route selection.
- Built chatbot-assisted navigation with rerouting, route severity analysis, and interactive map visualization using graph algorithms and spatial data structures.

Education

FAST National University of Computer and Emerging Sciences — Karachi, Pakistan
Bachelor of Science in Computer Science (BSCS) — CGPA: 3.16/4.00 — *Aug. 2024 – Expected May 2028*